

Myanmar Earthquake & Dam Risk Report

1. Executive Summary

This report covers 9,714 earthquake events in Myanmar from 1970-03-10 to 2026-07-21. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 1.037 with magnitude of completeness $M_c=4.7$.

Total Events: 9,714

Date Range: 1970-03-10 to 2026-07-21

Max Magnitude: M7.7

Mean Depth: 25.7 km

b-value: 1.037

a-value: 7.690

M_c (completeness): 4.7

2. Earthquake Map

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3. Seismic Clustering Analysis

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4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

Critical Risk: 44 dams

High Risk: 116 dams

Moderate Risk: 45 dams

Low Risk: 49 dams

Top 30 Highest Risk Dams:

Name	River	PGA(g)	Dist	Fault	Risk	Grade			
Dam near 21.74N 95.85E		nan	0.2718	7.4 km	10.0	Critical			
Dam near 22.09N 95.86E		nan	0.2995	5.7 km	10.0	Critical			
Dam near 21.77N 95.89E		nan	0.3257	2.7 km	10.0	Critical			
Dam near 21.47N 95.87E		nan	0.2815	6.0 km	9.7	Critical			
Dam near 20.85N 96.03E		nan	0.4813	8.3 km	8.8	Critical			
Meikhtila	nan	0.2412	8.2 km	8.8	Critical				
Dam near 21.34N 95.81E		nan	0.2263	12.2 km	8.8	Critical			
Natkar	nan	0.2443	7.9 km	8.7	Critical				
Dam near 21.37N 95.80E		nan	0.2207	13.4 km	8.7	Critical			
Dam near 20.72N 95.90E		nan	0.2665	5.6 km	8.7	Critical			
Dam near 21.04N 95.83E		nan	0.2270	10.9 km	8.7	Critical			
Dam near 21.05N 95.83E		nan	0.2265	11.0 km	8.7	Critical			
Dam near 20.47N 96.08E		nan	0.5000	9.9 km	8.6	Critical			
Dam near 20.60N 95.90E		nan	0.2464	7.4 km	8.6	Critical			
Taungnyo	Taungnyo	0.3054	1.2 km	8.6	Critical				
Nat Thar Taw	Nat Thar Taw	0.1903	20.2 km	8.5	Critical				
Myotha	Ku Htaw	0.1849	20.6 km	8.5	Critical				
Thaphan Chaung	Thaphan Chaung	0.2197	11.5 km	8.2	Critical				
Zeedaw	Nyaung Oat	0.2075	15.3 km	8.2	Critical				
Dam near 17.55N 96.37E		nan	0.1022	9.6 km	8.2	Critical			
Male-Nattaung	Male-Nattaung	0.2360	18.6 km	8.1	Critical				
Dam near 20.73N 95.83E		nan	0.2110	13.6 km	7.9	Critical			
Nyaunggone	nan	0.2104	13.7 km	7.9	Critical				
Paunggadaw	Paunggadaw	0.1488	33.9 km	7.9	Critical				
Kun Chaung	Kun Chaung	0.4644	0.9 km	7.7	Critical				
Kintat	Mu	0.1669	21.0 km	7.7	Critical				
North Pinle	Pinle	0.1427	36.1 km	7.7	Critical				
Phyu Chaung	Pyu Chaung	0.4707	1.3 km	7.6	Critical				
South Pinle	Pinle	0.1383	38.1 km	7.6	Critical				
Kabaung	Kabaung Chaung	0.3335	9.2 km	7.6	Critical				

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5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

K (productivity): 34.92

c (time offset): 16.408 hours

p (decay exponent): 0.772

Expected M $\geq$ 3 (7d): 4

Expected M $\geq$ 3 (30d): 18

Expected M $\geq$ 3 (90d): 52

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Abrahamson & Silva (2008) NGA-West1 GMPE
- Report generated by mmeq-opendata toolkit